Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

THE EVOLUTION OF MACHINE LEARNING AND AI

Week 5 Quiz Summary

Teaching a Machine a Law of Nature: everything the quiz can ask, in plain English

How to use this page. Week 5 is examined in the **Week 6 quiz** (50% of the module). The quiz is **multiple choice**. Everything it can ask about Week 5 is in the **ten points** below, in the order of the lecture.

Read the boxes (about 20 minutes). Do the questions (about 20 minutes). Mark them. Re-read only the points you got wrong.

If you remember one thing. Tell a machine the rule, and it needs no examples. Tell it the wrong rule, and it is wrong without warning: its own score stays small while its answers go wrong.

1 Part 1. The ten points

1.1 A pattern nudges; a law rules things out

A **pattern** is usually true in the examples you have: it nudges you towards likely answers and rules nothing out. A **law** is always true, everywhere: it removes whole sets of answers for good. Last week the guess built into a machine was a pattern ("nearby pixels usually belong together"); this week it is a law of nature. A law is a **question you can put an answer back into**, like marking homework without the answer sheet: 4 × 4 = 16, not 9, so 4 is wrong, and you never needed to know the answer was 3.

1.2 Told the rule, shown nothing, it won

The week's one rule is a weight on a spring: heaviness × change of speed + friction × speed + stiffness × position = 0, at every moment. Two copies of the same network, same training. One is checked against **64 measurements**: error **0.026**, levelling off. The other is shown **no measurements at all** and only checked against the rule: error **0.00135**, about **20 times better**. Where you will work, data is scarce (a new wind turbine, a rare condition, one bridge) and more cannot be bought; what you already know is the strongest help you have.

1.3 The leftover and the score

Put a guess back into the rule: whatever is left over instead of zero is the **leftover** (the **residual**); zero means the guess obeys. The machine's score (the **physics loss**) is the leftover, **squared and averaged** over a few hundred **check points** (**collocation points**). Check points cost nothing, because checking needs no answer, so they can go anywhere, even where no sensor could. **The true answer appears nowhere in the score.** The rule marks the homework: the true motion leaves nothing, a guess that forgets friction leaves a little, a flat line leaves a lot.

1.4 Only the check changed, and the slope of a slope

Week 3′s loop is guess, check, blame, nudge. Only the **check** changed: against the rule instead of against examples. A **physics-informed network** is an **ordinary network with a different check**; there is no special architecture. The one new trick: the computer must take the **slope of a slope** (how fast the speed changes) **exactly**. Nudging the input a tiny amount and dividing is nearly good enough for one slope and not good enough for a slope of a slope.

1.5 Beyond the data, it keeps going

Both models get measurements for the first part of the time only. Beyond the data, the examples-only model got **27.6 times worse**; the model told the rule was **no worse at all (1.0)**, because the rule was true out there too and it was checked there. A model that only saw examples **fails quietly** beyond them: smoothly, confidently, with no warning. **It does not know that it does not know.** Copy the start of a throw and the ball rises for ever; know gravity and it lands.

1.6 Energy for free, and the memoriser test

With no friction, the spring's energy must stay constant. **Energy was in nobody's score**, yet the examples-only model's energy drifted **16.3%**, the model with examples and the rule **5.1%**, and the rule-only model **0.2%**. A model that obeys the rule **inherits everything the rule implies**, whether or not you asked. The habit: **check something the rule implies that you never trained on**. It is how you tell a memoriser from an understander: ask a question that was not on the practice sheet.

1.7 The start, and a weight nobody can choose

The rule alone does not pick the motion: a weight that never moves also obeys it (0 + 0 + 0 = 0). **The start** (where it begins, how fast it is moving) does. Asking for the start as a second wish with a **weight** (a **soft penalty**) gave errors from **0.0015** to **0.66** depending on the weight, and **every version started in the right place**: the bad ones got the start perfect and neglected the rule, like a builder told a red door matters 500 times more than the house standing up. Nobody can choose the weight in advance. **Building the start into the sum** (a **hard constraint**: start + fade-in × network, with a fade-in that is zero and flat at the start) makes it always exact, with nothing to balance.

1.8 A small score proves less than you think; a wrong law is worse than none

Checked at **10** places, the score was the smallest of all and the real error the **largest** (0.30); at 50 and 200 places, small errors. **The score only answers "does it obey the rule where I checked?"** Then the most important result: told the friction is **four times** too high, the physics model was about **seven times worse than using no rule at all**, and **its own score stayed small**. **A wrong law is worse than no law, and the score will not tell you**: the confident expert with one out-of-date fact. You can only catch it against the real world.

1.9 Four more ways to build in science

Symmetry: if turning the input must not change the answer, let the model read only what turning cannot change (the distances): its answer changed **0.0%** under turning (the positions model **6.4%**), and it was no worse on unseen angles (**1.0** against **6.2** times). A **false symmetry is a wall**: with gravity added, the distances model cannot see which way is up and stays worse at **any amount of data**. **Learn the rule for the next step, not one route** (a **neural ODE**): three times past training, **1.7** times worse against **11.6**. Learn a **whole family** at once (a **neural operator**): a new case in a fraction of a millisecond, **faster but rougher**. **Picture generators** (diffusion) let physics dissolve a picture into noise and learn only how to undo one small step.

1.10 Choosing

Heat along a rod, from no data: the rule alone did about **40 times better** than 128 measurements. **Match the method to what you know**: a rule and one situation, a rule-checked network; thousands of situations, learn the family; **a rule on a simple shape, a classical solver**; an unknown number, run the network backwards; half the rule known, learn the next step and keep the known half; an exact symmetry, read only what cannot change; **no reliable rule, ordinary machine learning**. These methods are **not a better way to solve standard equations**; they are better for a little data plus a rule, unknown numbers, awkward shapes and fast stand-ins.

2 Part 2. The seven questions before you build

You may be asked to apply these, so read each one as a test you could run.

#	Question
1	What do I know, in plain words?
2	Is it exactly true, roughly true, or a guess?
3	Is it built into the sum, or asked for with a weight?
4	Where is the score checked, and where is it not?
5	What does the rule imply that I never trained on, so I can check it?
6	What happens if my rule is 20% wrong?
7	Would an older method, or ordinary machine learning, do this better?

3 Part 3. Twenty practice questions

Same format as the quiz: one right answer each. Answers on the last page.

1. Somebody tells you a person is in Dublin right now. Another tells you people in high-visibility jackets are often builders. Which statement captures the difference?

A. Both are patterns of different strength
B. The first is a law-like fact that rules answers out; the second is a pattern that only nudges
C. Both are laws, because both are usually true
D. The second is stronger, because it is about more people

2. How can you mark "find the number whose square is 9" without the answer sheet?

A. You cannot; you need the answer
B. By comparing with other students' answers
C. Put the answer back into the question and see whether it fits
D. By estimating roughly

3. The same network is trained twice. Shown 64 measurements, its error is 0.026. Shown none and checked only against the rule, its error is 0.00135. What was different?

A. Only what the guesses were checked against
B. The rule-checked network was much bigger
C. The rule-checked network trained for longer
D. The measurements were wrong

4. Why is this week's idea most useful where you are likely to work?

A. Because rules are cheaper to compute than data
B. Because neural networks cannot use measurements
C. Because classical solvers are obsolete
D. Because data is scarce there and more cannot be bought, while what you already know is free

5. What is the leftover (residual)?

A. The difference between the guess and a measurement
B. What remains when a guess is put back into the rule; zero means it obeys
C. The part of the data the model ignores
D. The error on the test set

6. Why are check points (collocation points) free?

A. Checking the rule needs no measurement, so they can be placed anywhere
B. They reuse the measurements already collected
C. They are only placed where sensors already exist
D. They do not affect the score

7. Is a physics-informed network a new kind of network?

A. Yes, it has a special physics architecture
B. Yes, it replaces gradient descent
C. No, it is an ordinary network with a different check; only that step of the loop changed
D. No, it is a classical solver

8. What is the one new capability the method needs?

A. A faster graphics card
B. More measurements
C. A bigger learning rate
D. Taking the slope of a slope exactly

9. Both models had measurements only for the first part of the time. Beyond it, the examples-only model got 27.6 times worse and the rule-checked model 1.0. Why did the second keep working?

A. It had seen more measurements
B. The rule was true beyond the data too, and it was checked there
C. It was trained for longer
D. Its errors happened to cancel

10. Why is "it does not know that it does not know" the dangerous property of the examples-only model?

A. It fails smoothly and confidently beyond its data, with no warning
B. It refuses to answer outside its data
C. It crashes when asked about new times
D. It always predicts zero beyond the data

11. Energy was in nobody's score, yet the rule-only model's energy drifted 0.2% against the examples-only model's 16.3%. What does that show?

A. Energy was secretly added to the loss
B. The rule-only model was lucky
C. A model that obeys the rule inherits everything the rule implies
D. Drift is always smaller with less data

12. What habit does the memoriser-and-understander comparison teach?

A. Train for longer on the same examples
B. Report only the training score
C. Add every known quantity to the loss
D. Check something the rule implies that you never trained on

13. A weight that never moves also obeys the spring's rule. What follows?

A. The rule alone does not pick the motion; the start must be given too
B. The rule is wrong
C. Every guess obeys the rule
D. The network cannot learn the rule

14. With the start asked for by a soft penalty, weights of 1, 50 and 500 gave errors of 0.0015, 0.32 and 0.66, and all three started in the right place. What went wrong with the bad ones?

A. They got the start wrong
B. They got the start perfect and neglected the rule, and nobody can choose the weight in advance
C. They needed more check points
D. The rule was wrong

15. What does building the start into the sum (a hard constraint) do?

A. Adds a third wish to the score
B. Requires choosing a larger weight
C. Makes the start exact whatever the network says, because the fade-in is zero at the start, leaving nothing to balance
D. Removes the need for the rule

16. Checked at 10 places, a model had the smallest score of all and the largest real error. Why?

A. The score was miscalculated
B. Ten places is always enough
C. The real error was measured wrongly
D. The score only says whether the rule is obeyed at the places checked; between them the answer wandered

17. Told the friction is four times too high, the physics model was about seven times worse than using no rule. Would its own score have warned you?

A. No: it obeyed the wrong rule well, so its score stayed small; only a check against the real world catches it
B. Yes: the score would have risen sharply
C. Yes: the model would have stopped training
D. No, but more check points would have revealed it

18. With gravity added, a model that reads only the distances between particles stays worse however much data it gets. Why?

A. It needs a larger learning rate
B. It was built unable to see which way is up, so a false symmetry is a wall
C. Distances are noisy
D. It was trained on too few angles

19. What is the difference between a neural ODE and a model that learns one motion?

A. The neural ODE needs no training
B. The neural ODE learns the answer at every time directly
C. The neural ODE learns the rule for the next step, so it can start anywhere and keeps working further out
D. There is no difference

20. You must solve a standard equation on a simple rectangle. What does Week 5 recommend?

A. A physics-informed network, because it needs no data
B. Ordinary machine learning
C. A neural operator
D. A classical solver: faster, more accurate, and with guarantees

4 Answers

#	Answer	Why
1	**B**	A pattern nudges and rules nothing out; a law removes whole sets of answers for good. Point 1.
2	**C**	A law is a question you can put an answer back into. Point 1.
3	**A**	Same network, same training; only the check changed, and the rule won by about 20 times. Points 2 and 4.
4	**D**	Forty readings, two hundred patients, one bridge: what you already know is the strongest help. Point 2.
5	**B**	Put the guess back into the rule; what is left instead of zero is the leftover. Point 3.
6	**A**	Checking needs no answer, so checks cost nothing and can go anywhere. Point 3.
7	**C**	Guess, blame and nudge are unchanged; only the check is against the rule. Point 4.
8	**D**	The rule needs how fast the speed changes; nudging and dividing is not good enough. Point 4.
9	**B**	It is not stretching examples further; it is following a law that holds everywhere. Point 5.
10	**A**	A model that only saw examples fails quietly beyond them. Point 5.
11	**C**	Steady energy is a consequence of the rule, so obeying the rule delivers it for free. Point 6.
12	**D**	Ask a question that was not on the practice sheet. Point 6.
13	**A**	0 + 0 + 0 = 0: only the start tells the two motions apart. Point 7.
14	**B**	The builder and the red door: perfectly following badly weighted instructions. Point 7.
15	**C**	start + fade-in × network: at the first moment the network is multiplied by zero. Point 7.
16	**D**	A small score proves less than you think. Point 8.
17	**A**	A wrong law is worse than no law, and the score will not tell you. Point 8.
18	**B**	A true symmetry is a free guarantee; a false one is a wall at any amount of data. Point 9.
19	**C**	A route is one journey; a map lets you start anywhere. Point 9.
20	**D**	These methods are not a better way to solve standard equations. Point 10.

Scored under 15? Re-read the points for the questions you missed, then try again tomorrow. The answers say which point each question came from.